

Is the COVID-19 Pandemic a Chance for Organic Food in Germany?

Evidence from Reddit

1,827

German-language comments

2019-2022

Observation period

2

Subreddit communities

This portfolio edition presents the verified study design, principal findings, interpretation limits, and reproducibility materials from the original term-paper source.

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Portfolio edition prepared from the archived Qmd source and verified project outputs

Executive summary



This project examines how organic food was discussed in two German-language Reddit communities before and during the COVID-19 period. It combines API-based data collection, German text preprocessing, keyword analysis, and lexicon-based sentiment scoring.

Research questions

- Can Reddit text analysis reveal patterns in attitudes toward organic food, including changes around the COVID-19 period?
- How did discussions of health, taste, environmental impact, and price develop between 2019 and 2022?

Principal findings

- The collected discussion volume rose from **118 comments in 2020** to **540 in 2021** and **1,121 in 2022**.
- Price was the most frequently matched study theme: **534 comments**, compared with 295 health, 232 taste, and 140 environmental matches.
- Among comments with at least one SentiWS match, the share classified as negative declined from **49% in 2020** to **44% in 2022**, while average annual sentiment remained below zero.

The evidence describes a keyword-matched Reddit corpus. It does not establish that the pandemic caused a population-level change in attitudes or purchasing behaviour.

Research contribution

I defined the collection keywords, gathered and prepared the Reddit data in R, developed the German-language text-processing workflow, applied the sentiment lexicon, compared discussion themes over time, and wrote the original term paper.

Research design

The study used public comments from r/de and r/FragReddit collected for 2019-2022 through organic-food keyword searches. The workflow combined collection, cleaning, tokenisation, thematic keyword matching, and German-language sentiment scoring.

1. Collect PSAWR and Pushshift keyword searches	2. Prepare Deduplicate, clean, and tokenize German text	3. Analyse Theme matching and SentiWS scoring	4. Interpret Compare years and state evidence limits
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Analytical scope

- Corpus: 1,827 comments across four calendar years.
- Communities: r/de and r/FragReddit.
- Theme groups: health, taste, environment, and price.
- Sentiment method: SentiWS lexicon scores aggregated within comments.

Year	r/de	r/FragReddit	Total
2019	43	5	48
2020	106	12	118
2021	392	148	540
2022	589	532	1,121

Table 1. Collected comments by year and subreddit.

The number of comments is a property of the collection design and keyword query. It should not be interpreted as a direct measure of market demand.

Result 1: discussion volume

The collected discussion volume rose sharply after 2020, from 118 comments in 2020 to 540 in 2021 and 1,121 in 2022. Prior to 2021, most comments in the corpus came from r/de; by 2022, the counts from the two communities were similar.

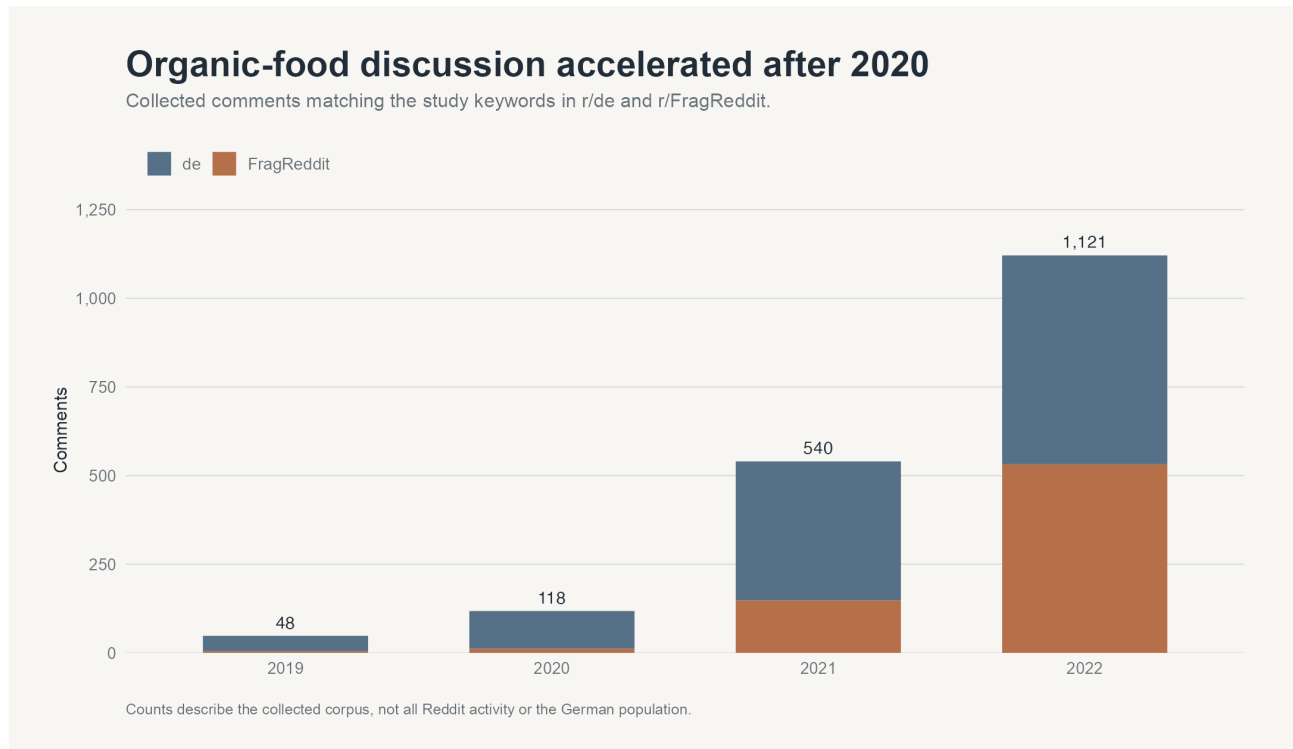


Figure 1. Annual comment counts in the collected corpus, split by subreddit.

The increase supports a finding about the visibility of organic-food discussion within this collected corpus. It does not prove that overall German consumer interest increased by the same amount.

Result 2: discussion themes

Price was the most frequently matched of the four study themes. Across the full corpus, price-related terms appeared in 534 comments, compared with 295 health matches, 232 taste matches, and 140 environmental matches. Health-related discussion became more visible after 2020, while environmental terms remained the least frequently matched group.

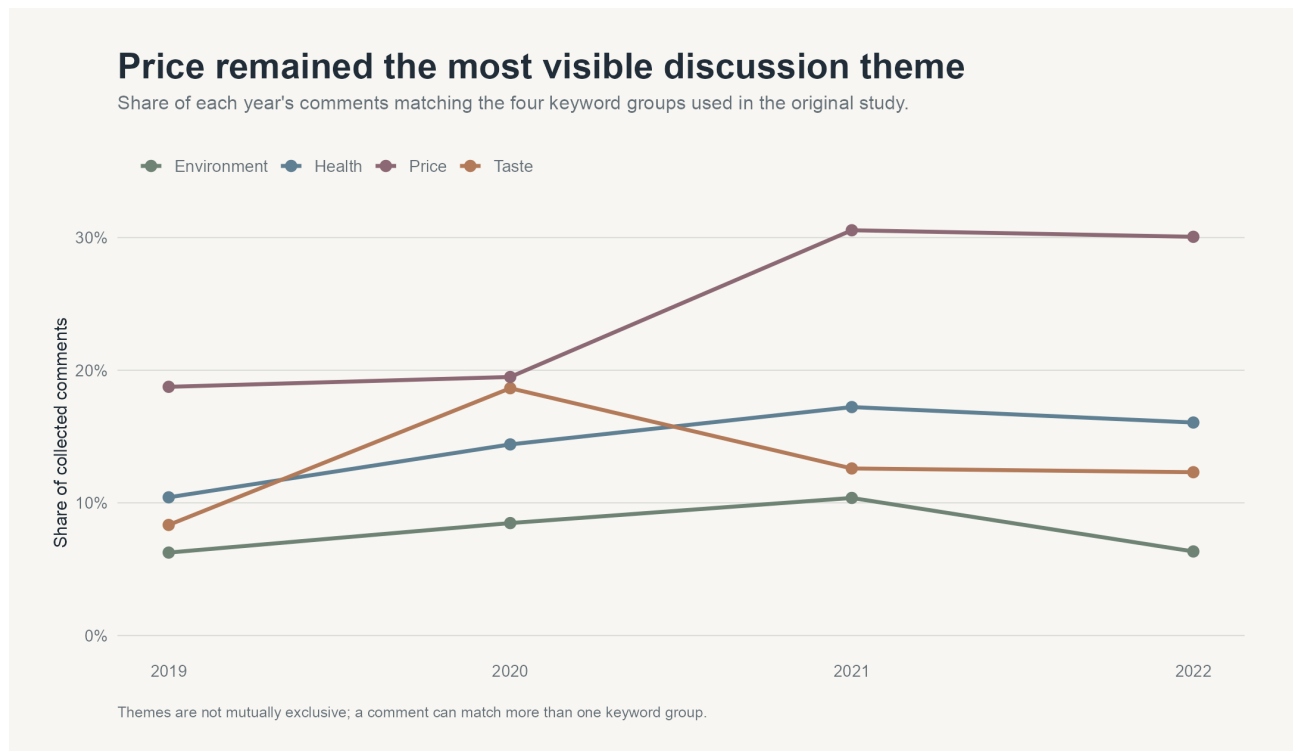


Figure 2. Annual share of collected comments matching each keyword group. Groups are not mutually exclusive.

Theme	Comments matched
Price	534
Health	295
Taste	232
Environment	140

Result 3: sentiment patterns

Each detected SentiWS term contributed a polarity score. Comment-level scores were calculated by summing matched terms; comments with scores below zero were classified as negative. The proportion classified as negative declined over time, but average annual scores remained below zero.

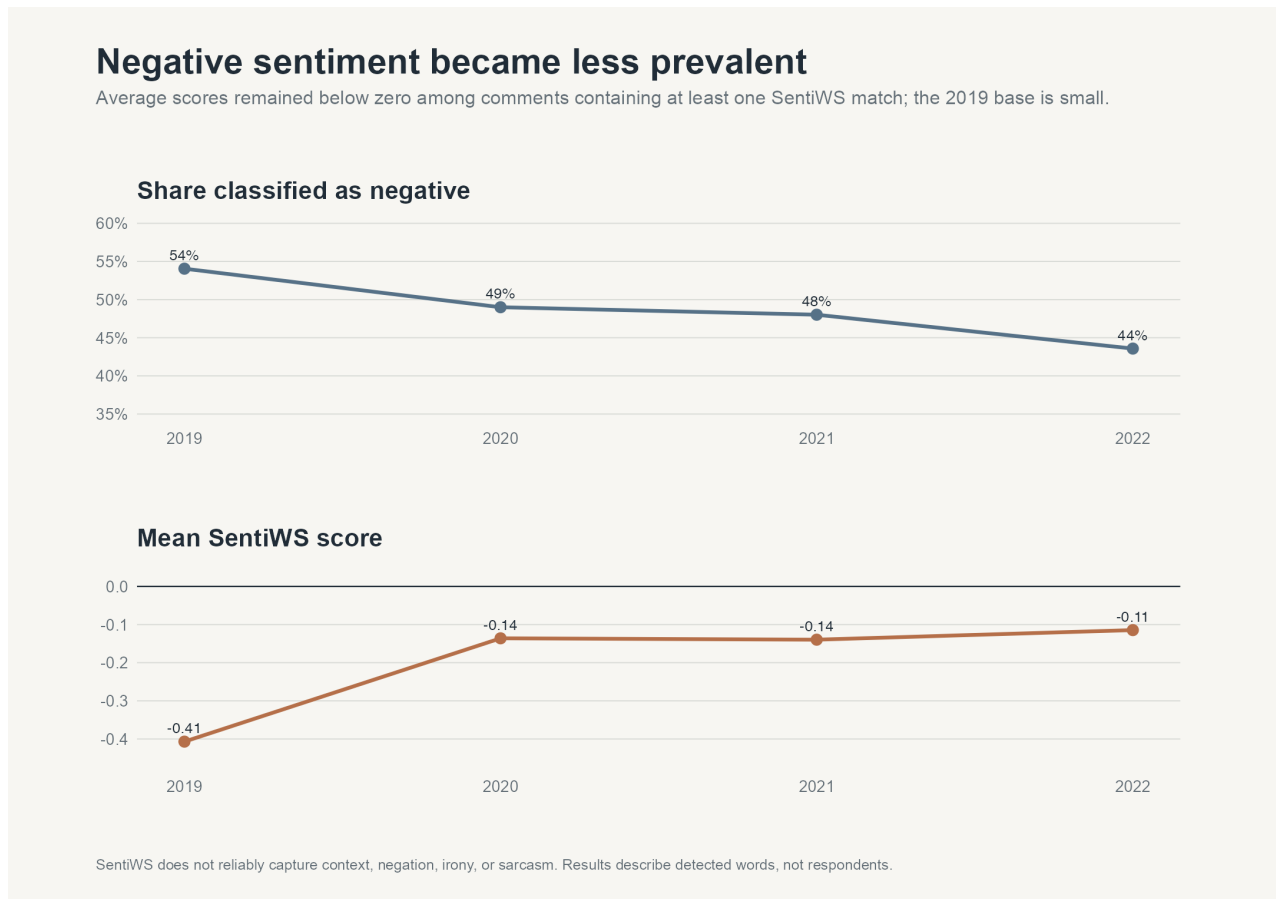


Figure 3. Negative classification share and mean SentiWS score among comments with at least one lexicon match.

Year	Comments scored	Mean score	Negative share
2019	37	-0.407	54%
2020	100	-0.136	49%
2021	429	-0.140	48%
2022	911	-0.114	44%

Interpretation and limits



Interpretation

The results point to a discussion shaped strongly by price and personal product experience. Health gained visibility during the pandemic period, while environmental language appeared less often. The sentiment trend suggests that negative language became less prevalent within the scored subset, without demonstrating a positive overall polarity.

Because the design observes keyword-matched comments rather than the same users over time, the findings do not establish that the pandemic caused a change in public opinion. Consumer expression, purchase intention, and observed purchasing behaviour represent different forms of evidence.

Limitations

- The data come from two subreddits and do not represent the German population.
- User nationality and residence cannot be verified.
- The 2019 base is small, with 48 collected comments.
- Manual keyword groups may omit relevant language or include off-topic uses.
- Lexicon-based sentiment has limited sensitivity to German context, irony, sarcasm, and negation.



The report separates descriptive patterns from causal claims. The figures show what appeared in the collected corpus and how the selected measures changed; they do not infer individual behaviour.

Practical value

The workflow demonstrates research design, API-based data collection, unstructured-text preparation, German-language sentiment analysis, data visualisation, analytical judgement, and concise insight communication.

Reproducibility and sources

Repository materials

File	Purpose
analysis/Term-paper.qmd	Original term-paper narrative and analysis source.
analysis/Data-Collection.qmd	Original data-collection procedure and search terms.
analysis/build_readme_figures.R	Rebuilds the three verified visual summaries.
data/Dataset.csv	Archived 1,827-comment corpus.
data/derived/	Compact aggregates used for figures and verification.
data/README.md	Field definitions, provenance, ethics, and interpretation limits.

Method resources

- Reddit comments were collected in the original study through PSAWR and the Pushshift interface.
- German-language sentiment scoring uses SentiWS, available at <https://osf.io/x89wq/>.
- The original academic source remains available in Qmd format. The supplied archive did not include its separate references.bib file, so this portfolio edition cites the method resources directly and preserves the Qmd as the authoritative full source.

Responsible reuse

The corpus contains public comments but should still be handled carefully. It should not be used to identify, contact, profile, or quote individual users. Any redistribution or extension should follow current platform terms and relevant research-ethics requirements.

Suggested citation: Luo, Shengjia. *Is the COVID-19 Pandemic a Chance for Organic Food in Germany? Evidence from Reddit*. Term paper, Ludwig-Maximilians-Universität München, 2023.